Adroit Technologies Protocol Driver

Name	Lovato RGAM Serial
DLL	LOVRGAM

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1. Overview

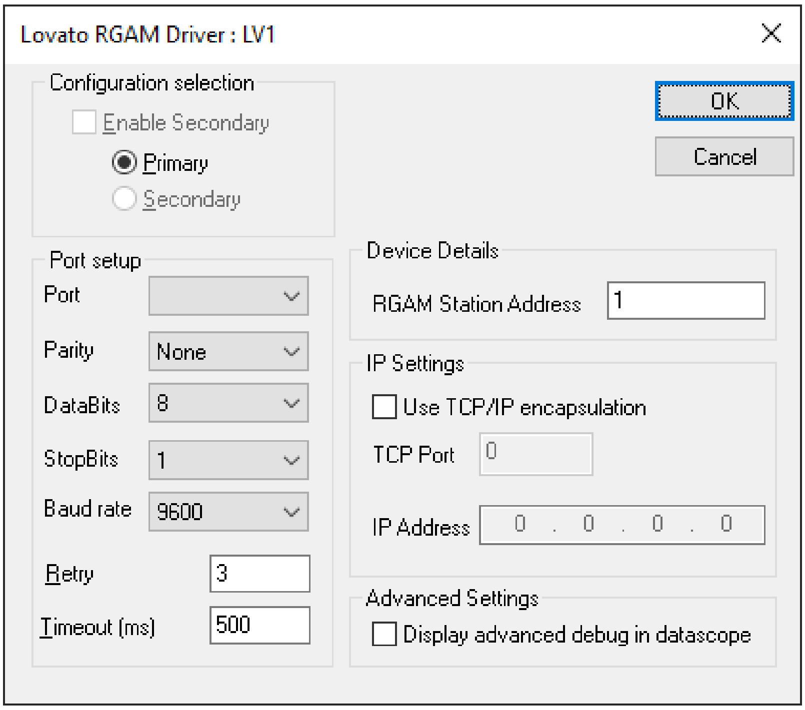
This document describes the setup of, and the communication protocol implemented in the Adroit protocol driver with the Lovato RGAM.

2. Wiring

Standard serial interconnection (RS232-C).

3. Device Configuration

3.1. Device Configuration



3.1.1. Configuration Selection [not implemented]

Enable Secondary – (not available) Enables secondary configuration for dual redundant Controllers. This setting is disabled for this driver.

Primary/Secondary – Select the primary or secondary button depending on which connection you wish to configure.

3.1.2. Port Setup

Port – Select the serial port to use for communications.

Parity – Select even or uneven parity in a single serial transmission message.

DataBits – Select the number of data-bits per packet.

StopBits – Select the number of stop-bits per packet.

Baud Rate – The transmission rate of data (including start and stop bits). The default baud rate for an RGAM unit is 9600.

Retry – The number of times to retry a transaction before a comms-failure is reported.

Timeout – The number of milliseconds to wait between communication-retries.

3.1.3. Device Details

RGAM Station Address – Supply the unit address of the RGAM device to communicate with.

3.1.4. IP Settings

If an Ethernet gateway is used to provide an encapsulated Serial interface to the controller, the following settings is used to configure the connection:

Use TCP/IP encapsulation – Ticking this box switches the driver from communicating using a COM Port to using a TCP/IP Socket.

TCP Port – The TCP Port on which the driver should try to connect to the controller.

IP Address – The IP Address that will be used for communicating with the controller.

3.1.5. Advanced Settings

This checkbox is available so that during setup the values being sent to and read from the device are displayed. The breakdown of what is sent back to Adroit is displayed and any errors that occur during transmission are displayed.

Display advanced debug in datascope – Displays extra information for debugging purposes during setup.

4. Supported Addresses

The Lovato RGAM Adroit serial driver supports the following read-only data address types:

Op-code	Description	Min Addr	Max Addr	Addressing example	Supported Adroit Datatypes
ST	General status request	1	12	ST 1 ST 12	Digital
		1	1	ST 1	Integer
ER	Error status request	1	21	ER 1 ER 21	Digital
		1	2	ER 1 ER 2	Integer
MV	Mains voltage value (3-phase) request	1	3	MV 1 MV 3	Integer Real
GV	Generator voltage value request	1	1	GV 1	Integer Real
FR	Generator frequency request	1	1	FR 1	Integer Real
BV	Battery voltage request	1	1	BV 1	Real
HR	Engine total working time request (minutes)	1	1	HR 1	Integer
MN	Time-to-maintenance request	1	1	MN 1	Integer
AV	Battery charger alternator voltage request	1	1	AV 1	Integer Real
IO	Digital input / output status request	1	15	IO 1 IO 15	Digital
		1	1	IO 1	Integer
CW	Control Bits for simulation of the buttons on the controller	1	12	CW 1 CW 11	Digital

5. Protocol Definition and Considerations

5.1. Lovato RGAM Message Structure

Message start	RGAM Station Address	Op-code	Operator	Data	Start Checksum	Check sum	Message Stop	Carriage return linefeed
(	01	ST	?	DD…	&	CK	)	\r\n

Message start:
The opening bracket character (0x28) indicates the start of a new message.

RGAM Station Address:
This two-digit number indicates the address of the RGAM to which the message has to be sent.

Op-code:
The op-code identifies the type of information that is requested by the user. This could be any one of the opcodes described in the previous section (Supported addresses).

Operator:
The operator indicates the type of message to send (Read or Write message). The Adroit Lovato RGAM serial driver only supports 'Read' messages; thus, the operator will always be a question mark ('**?**').

Data:
The data that the user wants to write to the unit. Because this driver doesn't support write functionality, the data portion will always be omitted.

Start checksum:
The ampersand character ('**&**') indicates the start of the message checksum.

Checksum:
The checksum is two characters wide and is the lowest significant byte of the algebraic sum of the single bytes that are transmitted, starting from the RGAM station address to the last data byte.

Message stop:
The closing bracket ('**)**') indicates the end of the message.

Carriage-return-linefeed:
The carriage-return and linefeed characters are appended to the end of each message.

5.2. General Status Request

The general status request message returns up to 12 bits of information, of which each of the bits indicates a certain named status within the unit. The different bits represent the following information:

Request	Status
ST 1	Bit (1) = TRUE if system is in OFF mode
ST 2	Bit (2) = TRUE if system is in MAN mode
ST 3	Bit (3) = TRUE if system is in AUT mode
ST 4	Bit (4) = TRUE if system is in TEST mode
ST 5	Bit (5) = TRUE if engine is running
ST 6	Bit (6) = TRUE if the alarm delay (oil + temp) has elapsed
ST 7	Bit (7) = TRUE if the mains voltage is present and within limits
ST 8	Bit (8) = TRUE if the generator voltage is present and within limits
ST 9	Bit (9) = TRUE if the mains contactor is closed
ST 10	Bit (10) = TRUE if the generator contactor is closed
ST 11	Bit (11) = TRUE if there are live alarms
ST 12	Bit (12) = TRUE if Automatic test is enabled

5.3. Error Status Request

Any error status ranging from 1 to 17 can be checked if it is ON by entering the 'ER' prefix and the error status number in the scan address field.
For example, to return error 5, you would enter ER 5.

Request	Error
ER 1	Bit (1) = TRUE if Low oil pressure alarm is active
ER 2	Bit (2) = TRUE if Engine overheating alarm is active
ER 3	Bit (3) = TRUE if Low fuel level alarm is active
ER 4	Bit (4) = TRUE if Starting failure alarm is active
ER 5	Bit (5) = TRUE if Low battery voltage alarm is active
ER 6	Bit (6) = TRUE if High battery voltage alarm is active
ER 7	Bit (7) = TRUE if Charger alternator failure alarm is active
ER 8	Bit (8) = TRUE if Generator voltage failure alarm is active
ER 9	Bit (9) = TRUE if Generator over frequency alarm is active
ER 10	Bit (10) = TRUE if Generator under frequency alarm is active
ER 11	Bit (11) = TRUE if Generator overload alarm is active
ER 12	Bit (12) = TRUE if Engine stop failure alarm is active
ER 13	Bit (13) = TRUE if Emergency stop alarm is active
ER 14	Bit (14) = TRUE if Unexpected stop alarm is active
ER 15	Bit (15) = TRUE if Maintenance requested alarm is active
ER 16	Bit (16) = TRUE if Mains contactor failure alarm is active
ER 17	Bit (17) = TRUE if Generator contactor failure alarm is active
ER 18	Bit (18) = TRUE if User alarm 1 is active
ER 18	Bit (19) = TRUE if User alarm 2 is active
ER 20	Bit (20) = TRUE if User alarm 3 is active
ER 21	Bit (21) = TRUE if User alarm 4 is active

5.4. Mains voltage (3-phase) Request

Each one of the 3 phases of voltages can be retrieved by entering 'MV' and the specific voltage you require, for ex. phase voltage 2 would be accessed by using MV 2.

5.5. Generator Voltage Value Request

The generator voltage value can be requested by using the address GV 1.

5.6. Generator Frequency Request

The generator frequency value can be requested by using the address FR 1.

5.7. Battery Voltage Request

The battery voltage can be requested by using the address BV 1.

5.8. Engine Total Working Time (minutes)

The engine total working time (in minutes) is retrieved from the unit by using the address HR 1.

5.9. Time-to-Maintenance Request (minutes)

The time-to-maintenance (in minutes) can be retrieved by using the address MN 1.

5.10. Battery Charger Alternator Voltage

The battery charger alternator voltage can be retrieved by using the address AV 1.

5.11. Digital Input / Output Status

The status of the I/O lines of the RGAM can be retrieved by the following addresses:

Request	Status
IO 1	Bit (1) = TRUE if the oil pressure input is active
IO 2	Bit (2) = TRUE if the high temperature input is active
IO 3	Bit (3) = TRUE if the fuel level input is active
IO 4	Bit (4) = TRUE if the emergency stop input is open (active)
IO 5	Bit (5) = TRUE if programmable input terminal 10 is active
IO 6	Bit (6) = TRUE if programmable input terminal 11 is active
IO 7	Bit (7) = TRUE if programmable input terminal 12 is active
IO 8	Bit (8) = TRUE if programmable input terminal 13 is active
IO 9	Bit (9) = TRUE if the mains contactor relay contact is closed
IO 10	Bit (10) = TRUE if the generator contactor relay contact is closed
IO 11	Bit (11) = TRUE if the fuel valve relay contact is closed
IO 12	Bit (12) = TRUE if the start relay contact is closed
IO 13	Bit (13) = TRUE if programmable relay terminal 18 is closed
IO 14	Bit (14) = TRUE if programmable relay terminal 16 is closed
IO 15	Bit (15) = TRUE if programmable relay terminal 20 is closed

5.12. Control Bits

Commands can be sent to the controller by simulating button presses on the front plate of the controller. Commands will only be sent on the rising edge of the digital control value. For this reason it is recommended that digital agents with pulse output configured is used for these scan addresses.

Request	Status
CW 1	Simulates pressing the "OFF" button
CW 2	Simulates pressing the "MAN" button
CW 3	Simulates pressing the "AUT" button
CW 4	Simulates pressing the "TEST" button
CW 5	Simulates pressing the "SELECT/RESET" button
CW 6	Simulates pressing the "START" button
CW 7	Simulates pressing the "STOP" button
CW 8	Simulates pressing the "MAINS" button
CW 9	Simulates pressing the "GEN" button
CW 10	Resets the maintenance interval
CW 11	Reboots the system.

6. General Notes and Troubleshooting

6.1. Contact Details

Email: support@adroit.co.za and drivers@adroit.co.za

Phone: +27 11 658-8100

Web: www.adroit.co.za

6.2. Adroit Technologies Knowledge Base

For additional information, please refer to our Knowledgebase website:

Latest Drivers topics - Features, discussions, tips, tricks, questions, problems and feedback (adroittechnologiesautomation.com)

7. Revisions

7.1. Document Revision

Rev.	Date	Author	Comment	Approval
1.0	14-Dec-2022	J. Duncan	New Format	D. Jordaan
1.1	25-Jan-2023	K. Robinson	Changes to format, updated image resources. Added Appendix B DCOM Hardening. Removed Windows XP section	D. Jordaan
2.0				
3.0				

7.2. DLL Revision

Rev.	Date	Changes
1	31-Jan-2005	First release notes! Beta Release Available
8	25-Mar-2014	Added cluster awareness.
9	28-Sep-2018	Perfmon counter were not available when running the Agent Server as a service.